Nuclear Fuel Performance

NE-533

Spring 2026

Last Time

- Radiation effects in UO2
- Can increase lattice parameter and decrease Young's modulus
- High burnup structure
  - higher power on periphery, higher burnup, lower temperatures
  - leads to grain refinement and large amounts of porosity
  - fission gas is retained, despite high porosity
  - thermal conductivity slightly increases due to 'clean' grains
- Fuel chemistry
  - different U charges can accommodate defects/substitutionals
  - O/M ratio is linked to oxygen partial pressure (oxygen potential)
  - O/M ratio is buffered by Mo in the fuel, and decreased near the cladding
  - stoichiometry affects fuel properties

Last Time

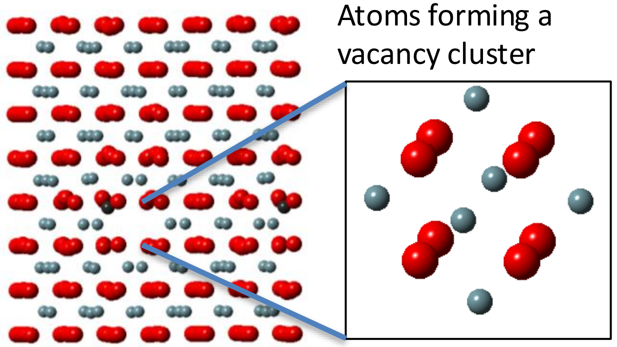
- Five families of fission products
  - soluble oxides (in fluorite structure)
  - insoluble oxides (gray oxide phase)
  - metals (white inclusions, 5-metal precipitates)
  - volatiles (gas bubbles)
  - noble gases (gas bubbles)
- Impact thermal conductivity and swelling in different ways
- Which fission products are in which group change based upon oxygen potential

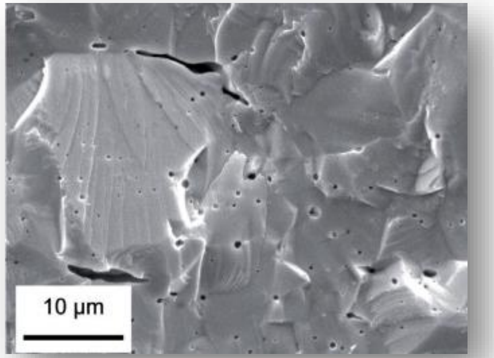
FISSION GAS RELEASE

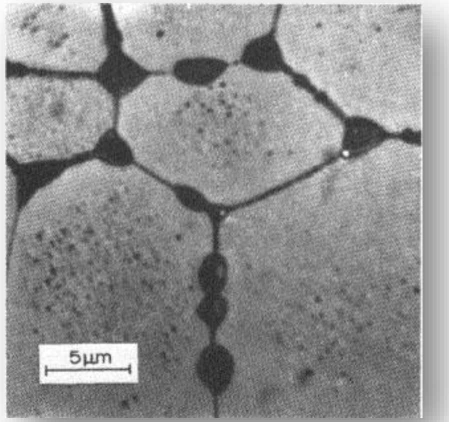
Fission Gas Release

- Fission gases (Xe, Kr) are released in a process composed of three stages in UO_2
- Stage 1: Gas atoms are produced throughout the fuel due to fission and diffuse towards grain boundaries
- Small intragranular bubbles form within the grains, but never get larger than a few nm radius due to resolution from energized particles
- Gas atoms that don't get trapped within the intragranular bubbles migrate to grain boundaries

Los Alamos NATIONAL LABORATORY EST. 1943

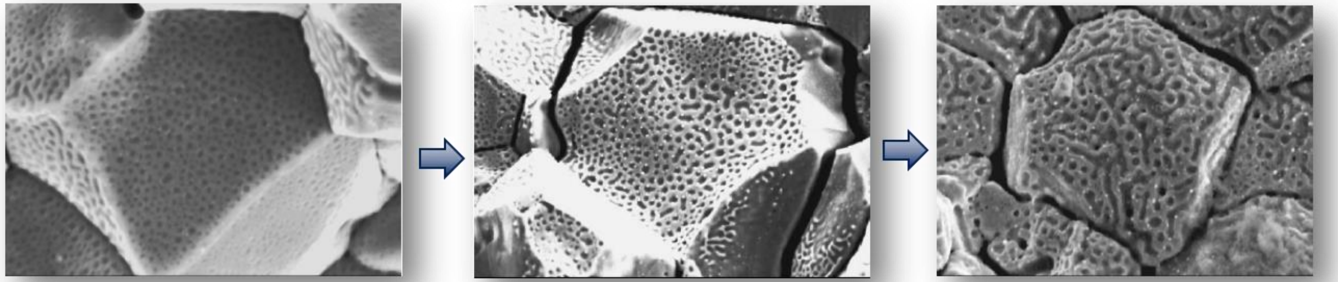






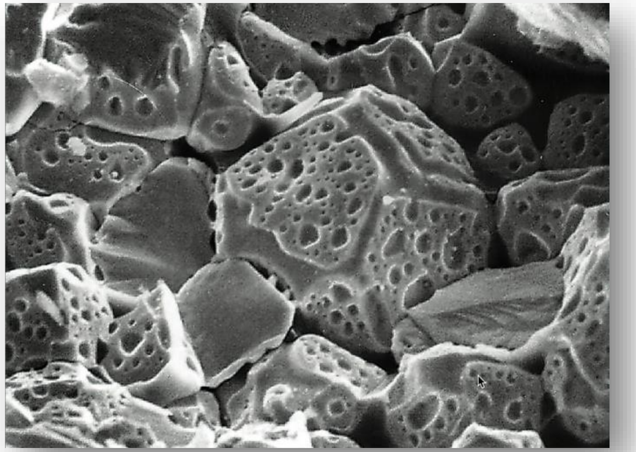
Fission Gas Release

- Stage 2: Gas bubbles nucleate on grain boundaries, growing and interconnecting



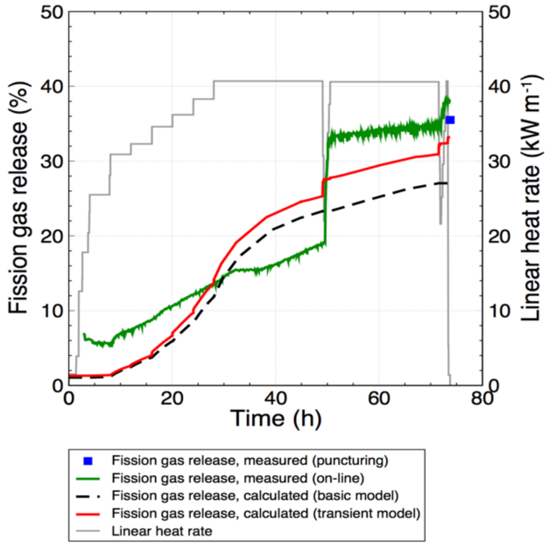
Nucleation Growth Percolation

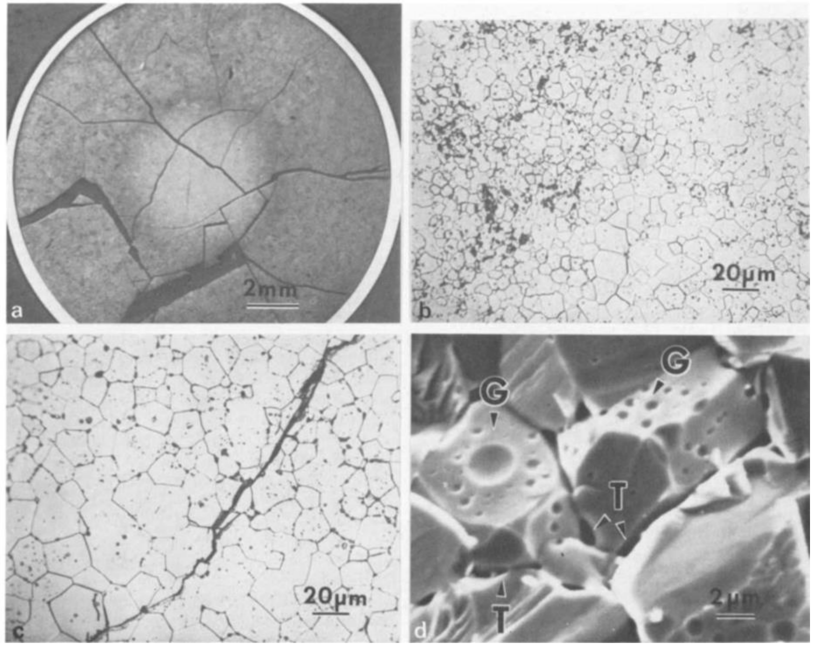
- Stage 3: Gas travels through interconnected bubbles to a free surface



Fission Gas Release

- Fission gas release also occurs due to mechanisms that don't depend on diffusion
- Release can occur to particle recoil and knockout at low temperature
- It can occur due to fracture during rapid transients





Fission Gas Release

- Released fission gas enters the gap and plenum, causing various problems
- Xe and Kr have very low thermal conductivities, reducing the gap conductance
- The plenum pressure increases
- The volatile fission gases corrode the cladding
- They are also radioactive and hazardous, causing problems when the cladding is breached
- Fission gas release experiments:
  - Post irradiation annealing
    - Fuel is irradiated at low temperature; Fuel is then placed into a furnace and heated; Gas atom release is then measured
  - In-pile release
    - Gas release is measured during reactor operation; It is much more difficult than post-irradiation annealing; Total amount released is measured by puncturing cladding after irradiation; Release with time can be estimated using a pressure transducer inside an instrumented fuel rod

Fission Gas Release

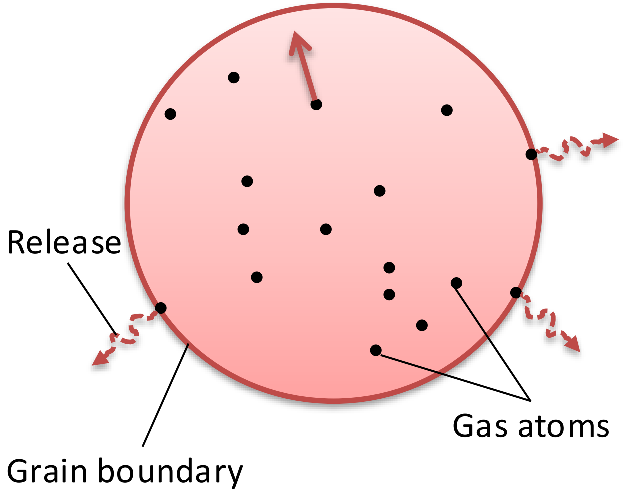
- Fission gas release models attempt to predict the rate at which gas is released from the fuel
- To model fission gas release, ideally, we must model all three stages of gas release
  - Diffusion of gas atoms to grain boundaries
  - Growth and interconnection of grain boundary bubbles
  - Transport of gas atoms through interconnected bubbles to free surfaces
- The earliest models only considered Stage 1
- Most models now consider stage 1 and 2
- There are no models that consider all three stages, but some are under development

Booth Model

- The Booth model is the earliest model of fission gas release and only considers stage 1
- A grain is considered as a simple sphere
- Gas atoms are released at the grain boundary
- The model solves the diffusion equation in 1D spherical coordinates
- Assumptions
  - $c_g(r, t)$
  - All grains are spheres of radius *a*
  - D is constant throughout the grain
  - Gas is produced uniformly throughout the grain
  - Gas is released once it reaches the grain boundary

$$\dot{c}_g = k_{c_g} + \nabla \cdot D \nabla c_g$$

$$\dot{c}_g = k_{c_g} + D \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_g}{\partial r} \right)$$



<u>ICs and BCs</u>

$c_g(r, 0) = 0$

$c_{g,r}(0, t) = 0$

$c_g(a, t) = 0$ (release)

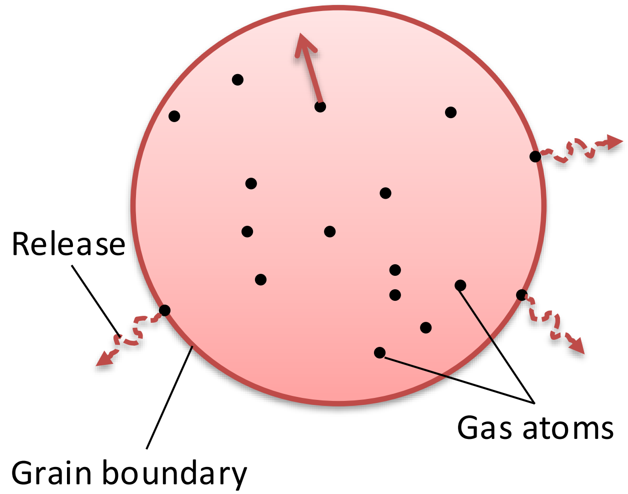
Modeling post-irradiation annealing

- The initial gas concentration is c_g^0
- No gas is produced

$$\dot{c}_g = D\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c_g}{\partial r}\right)$$

- Solving this equation tells us the value of c_g at any radius or time
- However, we want to know the fraction of gas atoms that have made it to the grain boundary
- We use the flux at the grain boundary

$$J_a = -D\left(\frac{\partial c_g}{\partial r}\right)_a \qquad f = \frac{4\pi a^2 \int_0^t J_a dt}{4/3\pi a^3 c_g^0} = \frac{3}{a c_g^0}\int_0^t J_a dt$$



<u>ICs and BCs</u>

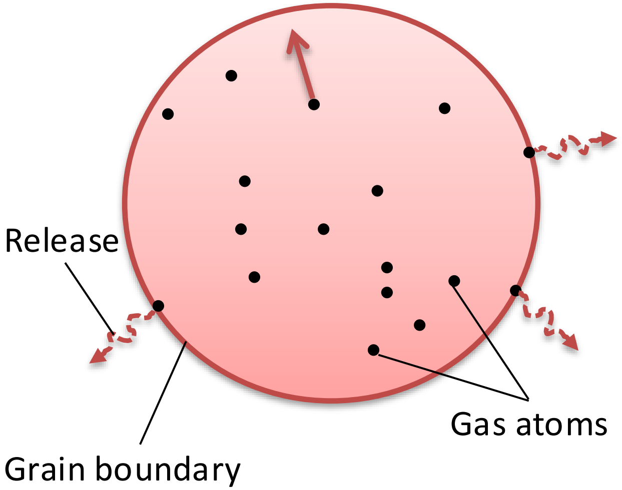
$c_g(r, 0) = c_g^0$

$c_{g,r}(0, t) = 0$

$c_g(a, t) = 0$ (release)

Solving the Booth Model

- This equation is solved using a Laplace transform after nondimensionalization
- Will not go through the derivation (shown in Olander)
- | = D x t / a^2



$$f = 6\sqrt{\frac{Dt}{\pi a^2}} - 3\frac{Dt}{a^2} \qquad \tau < \pi^{-2}$$

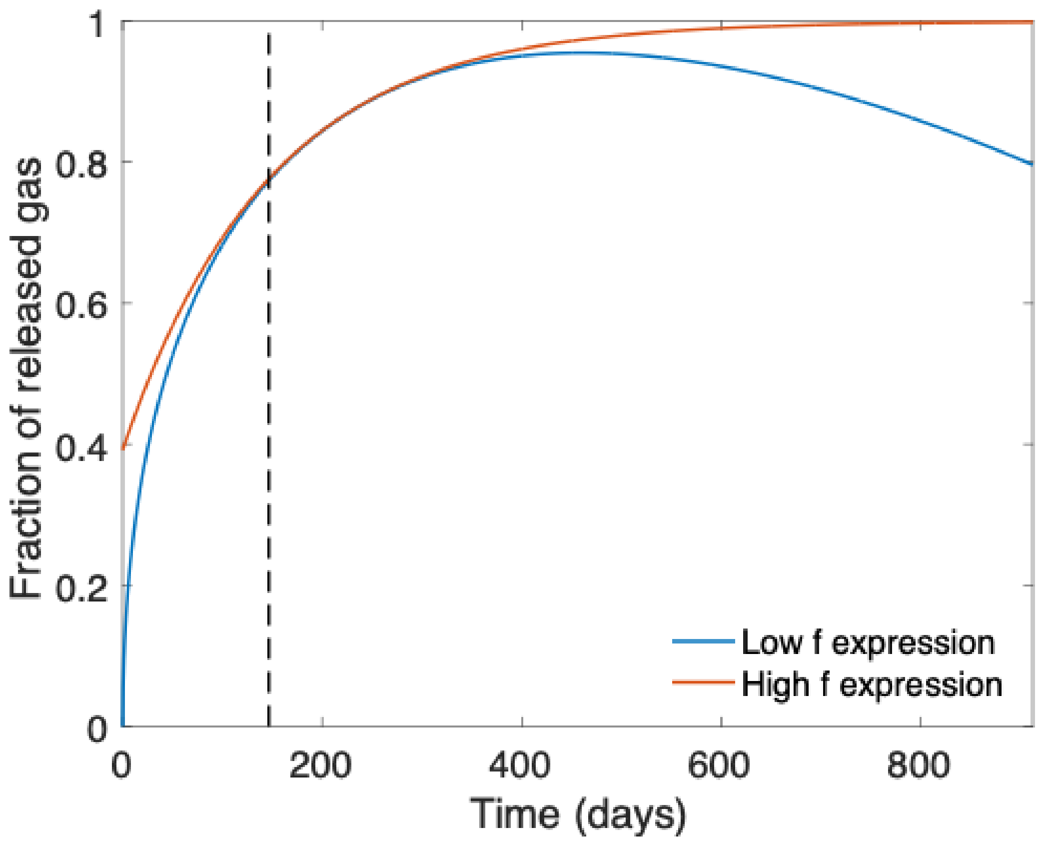
$$f = 1 - \frac{6}{\pi^2}e^{-\pi^2 \frac{Dt}{a^2}} \qquad \tau \geq \pi^{-2}$$

Booth Example

- For a diffusion coefficient for Xe of D = 8e-15 cm^2/s, what fraction of the fission gas trapped in a post-irradiation annealed fuel pellet has escaped after one hour? It has an average grain size of 10 microns
  - D = 8e-15 cm^2/s
  - a = 10e-4 cm
  - t = 3600 s
- Which f? | = D x t/a^2 = 2.88E-4 < π^{-2} = 0.101 $f = 6\sqrt{\frac{Dt}{\pi a^2}} - 3\frac{Dt}{a^2}$
  - f = 6*sqrt(8e-15*3600/(pi*(10e-4)^2)) – 3*8e-15*3600/(10e-4)^2 = 0.0181

Different expressions for fission gas release

- Given the data from the previous example, can plot both
  - $\tau < \pi^{-2}$ $$f = 6\sqrt{\frac{Dt}{\pi a^2}} - 3\frac{Dt}{a^2}$$
  - $\tau > \pi^{-2}$ $$f = 1 - \frac{6}{\pi^2} e^{-\pi^2 \frac{Dt}{a^2}}$$



Modeling in-pile release

- The initial gas concentration is 0
- Gas is produced due to fission, where y is the chain yield (y = 0.3017 for Xe and Kr) and the fission rate

$$\dot{F} = qN_U\sigma_{f235}\phi_{th}$$

- Gas can also decay, where λ is the decay constant
  – If we only consider stable stable products, λ = 0
- For in pile release, the fraction is equal to

$$f = \frac{3}{ay\dot{F}t}\int_0^t J_a\,dt$$

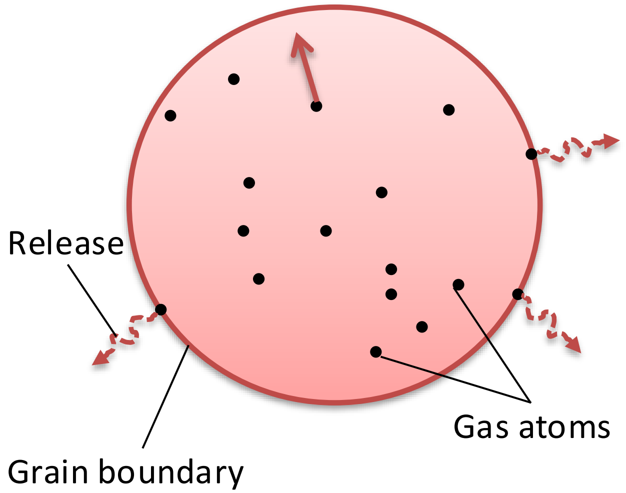
- After solving with with a Laplace transform

$$f = 4\sqrt{\frac{Dt}{\pi a^2}} - \frac{3}{2}\frac{Dt}{a^2} \qquad \tau < \pi^{-2}$$

$$f = 1 - \frac{0.0662}{\frac{Dt}{a^2}}\left(1 - 0.93e^{-\pi^2\frac{Dt}{a^2}}\right) \qquad \tau \geq \pi^{-2}$$

- The total gas production is $y\dot{F}t$ gas atoms/cm³

$$\dot{c}_g = y\dot{F} + D\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c_g}{\partial r}\right) - \lambda c_g$$



<u>ICs and BCs</u>

$c_g(r, 0) = 0$

$c_{g,r}(0, t) = 0$

$c_g(a, t) = 0$ (release)

Example

- For a diffusion coefficient for Xe of D = 8e-15 cm^2/s, what fraction of the fission gas trapped in an in-pile fuel pellet has escaped after one hour? It has an average grain size of 10 microns.
  - D = 8e-15 cm^2/s
  - a = 10e-4 cm

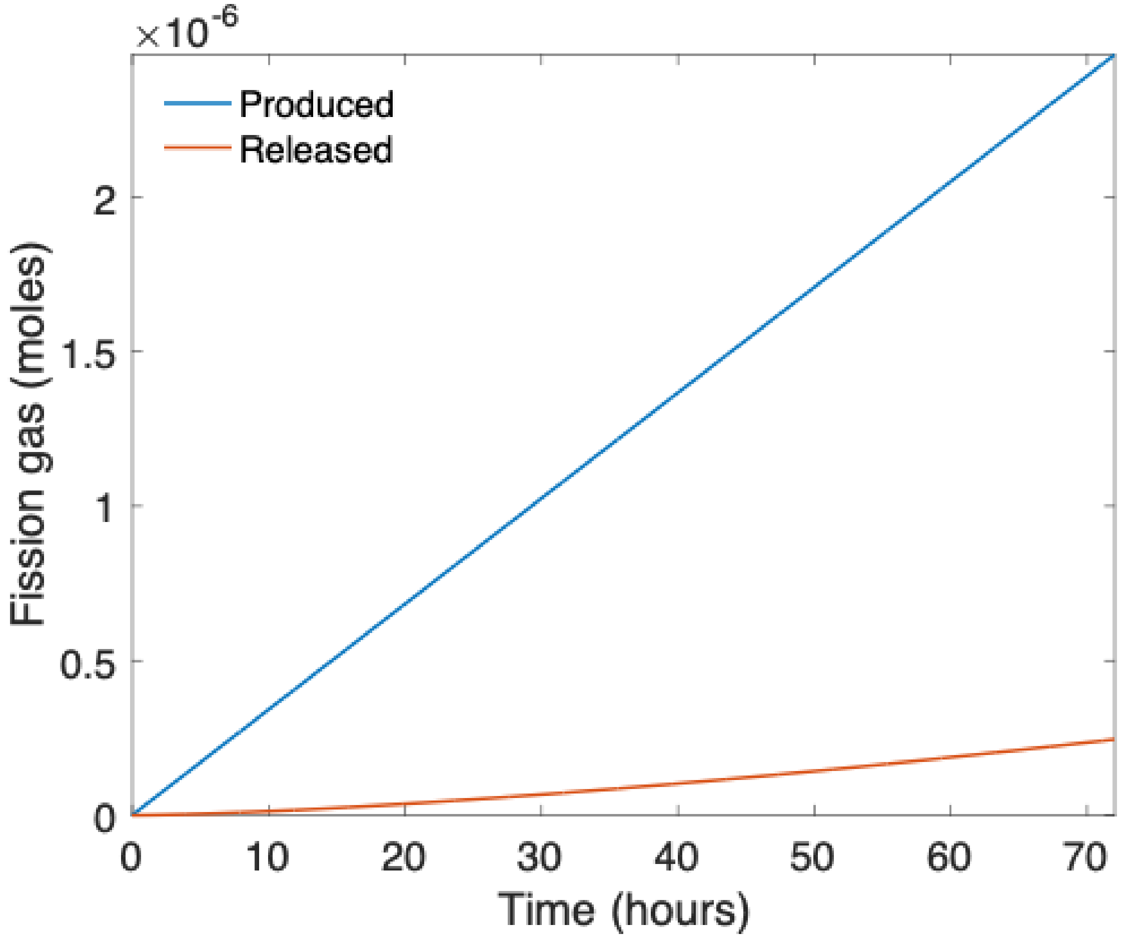
$$\tau = \frac{Dt}{a^2}$$

  - We have a short time (t=3600 s, $\tau < \pi^{-2}$), so we can use:

$$f = 4\sqrt{\frac{Dt}{\pi a^2}} - \frac{3}{2}\frac{Dt}{a^2}$$

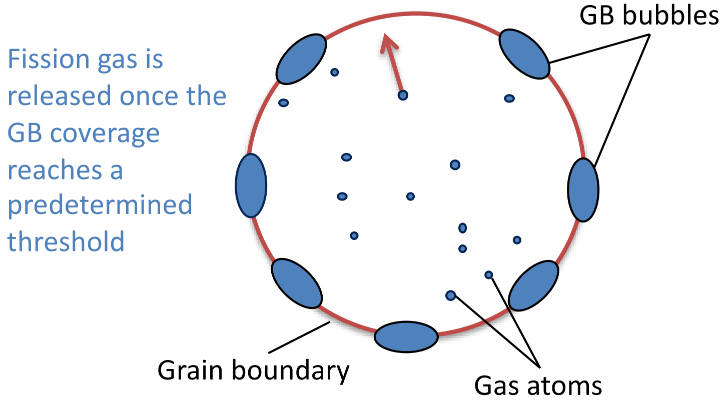
  - f = 4*sqrt(8e-15*3600/(pi*(10e-4)^2)) – 3/2*8e-15*3600/(10e-4)^2 = 0.0121

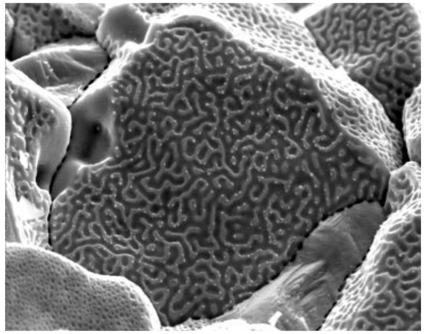
As time progresses, both the fraction released and the produced gas increase



Forsberg-Massih model

- The Booth model ONLY considers stage one of fission gas release
- Two stage Forsberg-Massih mechanistic model
  - Considers intragranular diffusion diffusion to grain boundaries (stage 1)
  - Also, grain boundary gas accumulation, resolution back into grain, saturation (stage 2)
  - Assumes that once the bubbles on the grain face are interconnected, it is released (no stage 3)





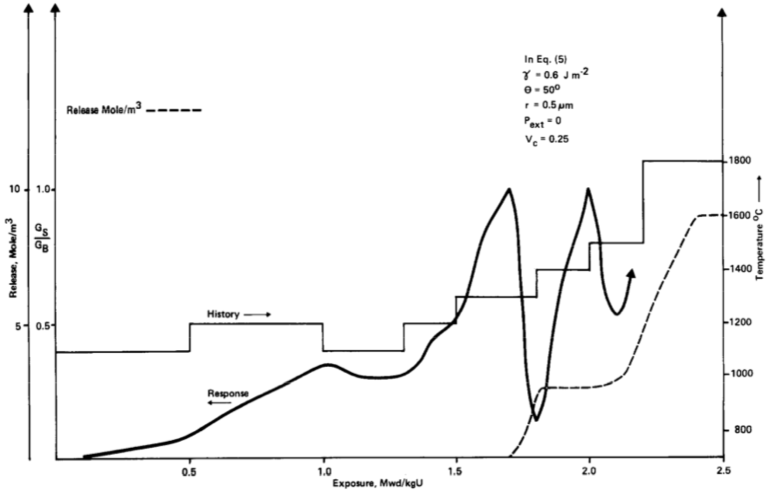
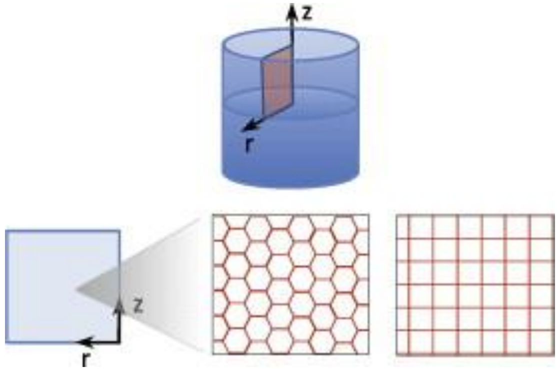
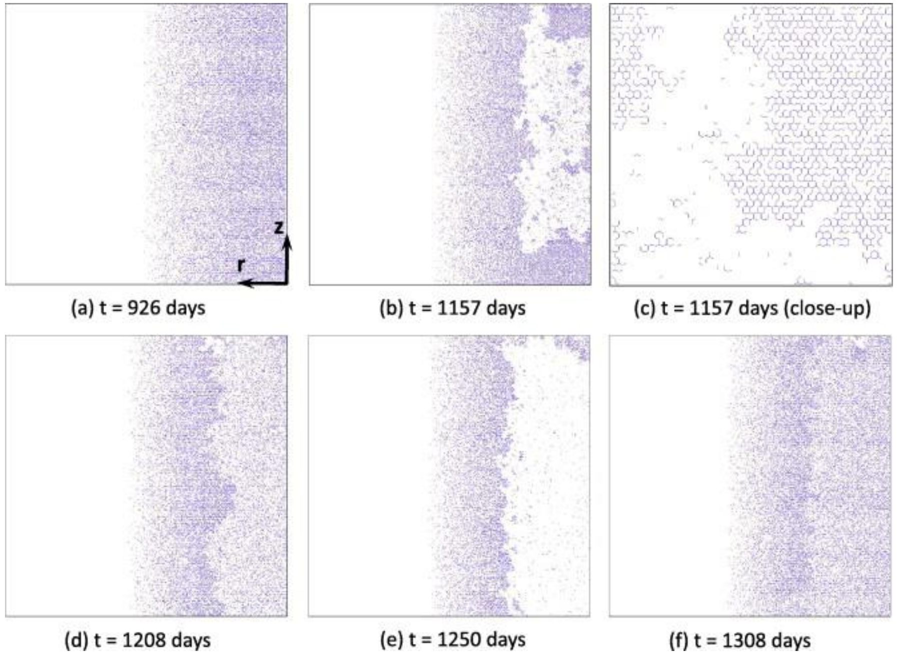


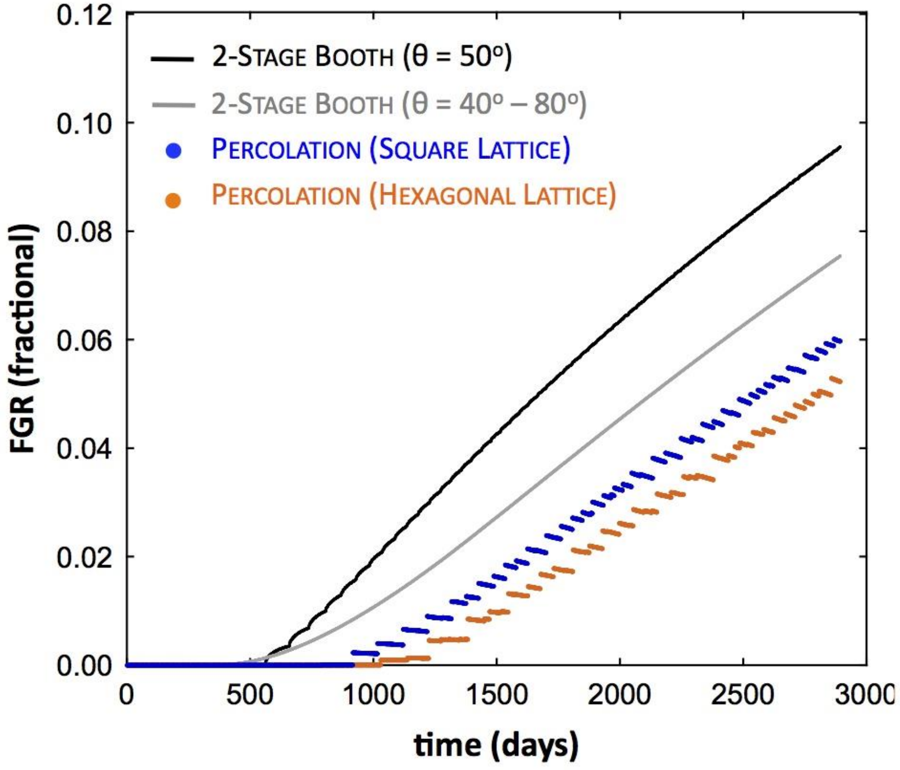
Fig. 1. Fraction of gas atoms on grain boundary, G_s/G_B, as a function of exposure for downward fuel cascading temperature history. γ is the bubble surface tension, 2θ is the angle where two free surfaces meet at a grain boundary, r is average bubble radius, V_c is the fractional coverage of the grain boundaries at saturation and the grain radius is taken to be 5 μm.

Forsberg-Massih model

- 2-stage F-M model over-predicts gas release because it neglects grain boundary bubble percolation (Stage 3)



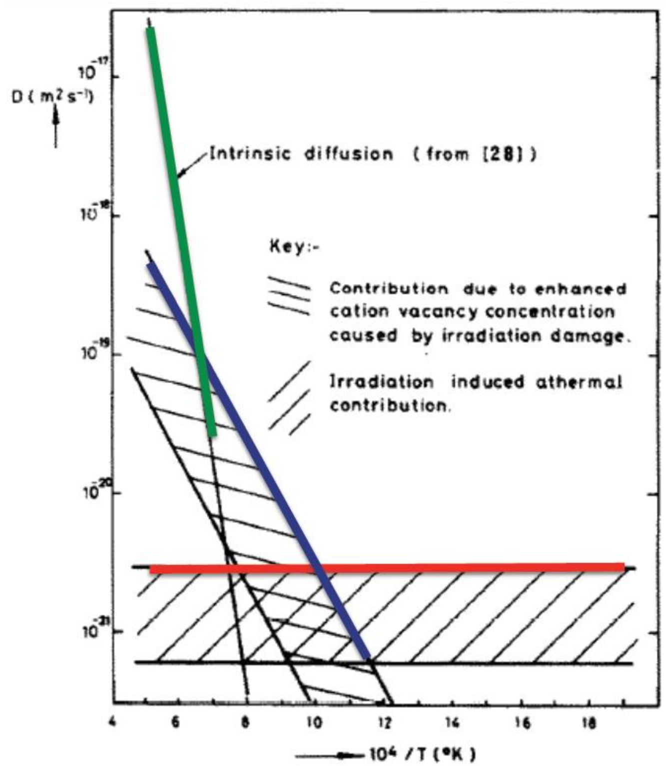




Millett, Tonks, and Biner. JNM, 424.1 (2012): 176-182.

Gas diffusion

- The diffusivity of the fission gas depends on temperature and on irradiation
- Experimental data shows three different regimes for the diffusivity

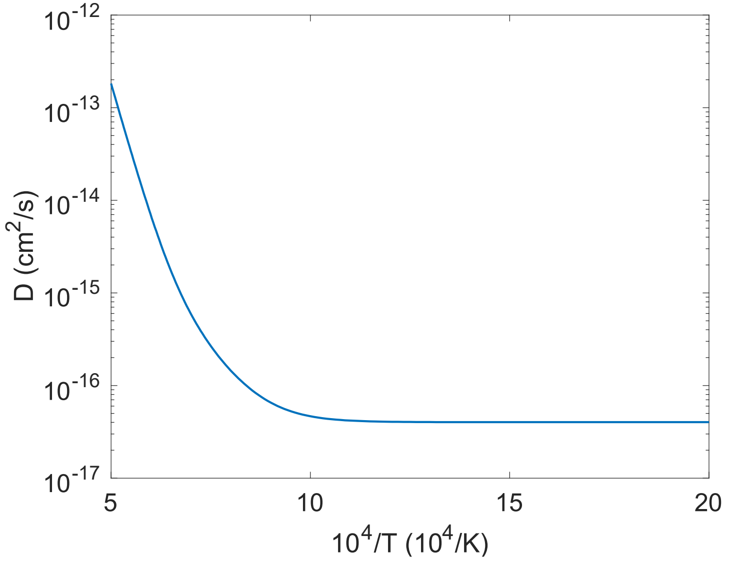


$$D = D_1 + D_2 + D_3 \text{ cm}^2/\text{s}$$

$$D_1 = 7.6 \times 10^{-6} e^{-\frac{3.03 \text{ eV}}{k_b T}}$$

$$D_2 = 1.41 \times 10^{-18} e^{-\frac{1.19 \text{ eV}}{k_b T}} \sqrt{\dot{F}}$$

$$D_3 = 2.0 \times 10^{-30} \dot{F}$$



Gas diffusion

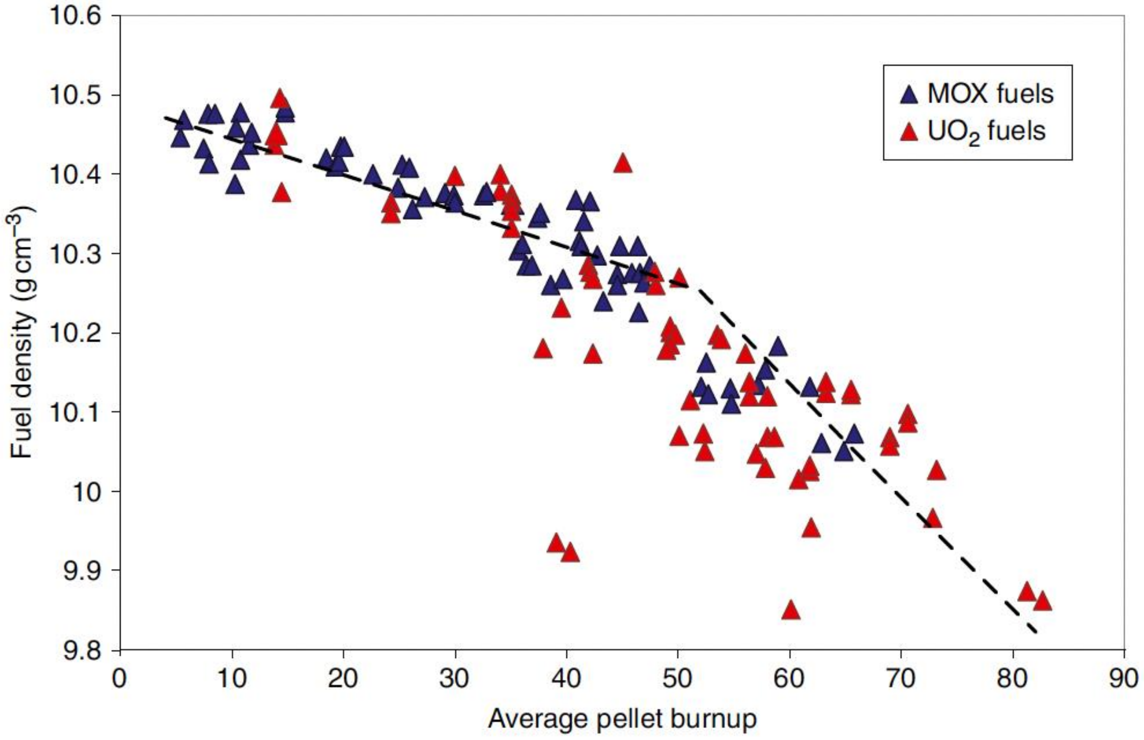
- The effective fission gas diffusivity is slower due to trapping by intragranular bubbles
- As the gas atoms diffuse towards the grain boundary, some are trapped by the small intragranular bubbles
- Some are later knocked out by energized particles (called resolution)
- The effective diffusion constant depends on the trapping rate r_t and the resolution rate r_r

$$D_{eff} = \left(\frac{r_r}{r_r + r_t}\right) D$$

FUEL SWELLING/DIMENSIONAL CHANGE

Fuel changes size and shape under reactor operation

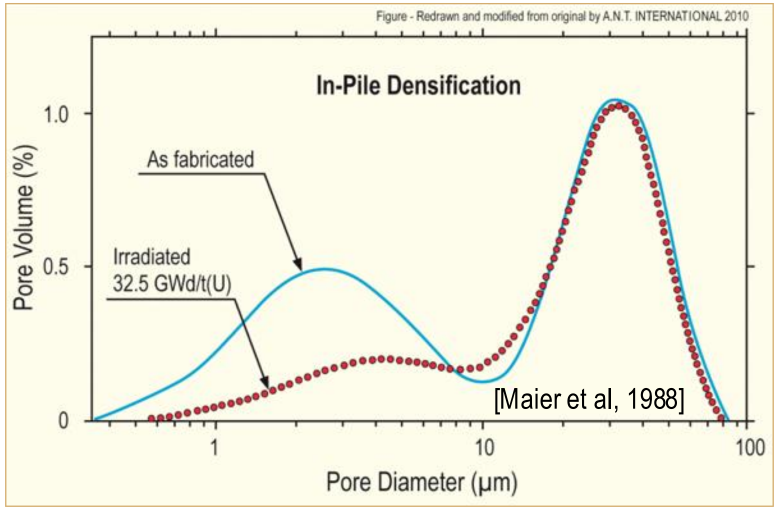
- Thermal expansion:
  - increase in volume, decrease in density, caused by increasing temperature
- Densification:
  - Decrease in volume, increase in density, caused by shrinking of porosity left after sintering
- Swelling:
  - Increase in volume, decrease in density, caused by fission products
- Irradiation Creep:
  - Change in shape, constant density, occurs with applied stress less than σ_y

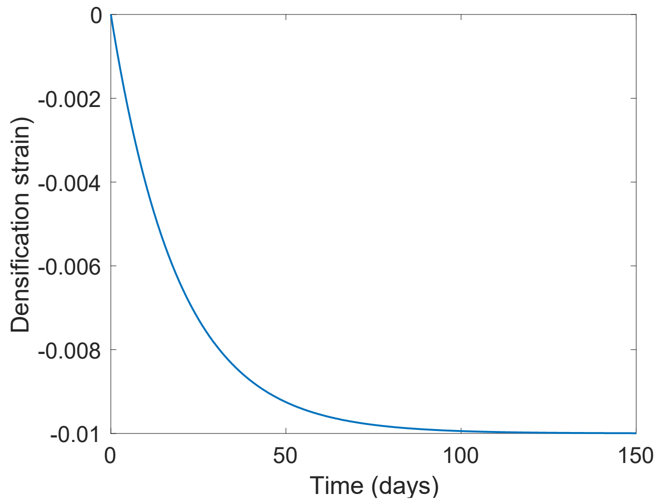


Densification

- Densification takes place during initial 5 - 10 MWd/kgU
  - Small, as-built pores close due to effects of fission spikes and vacancy diffusion
  - Large pores stable (in absence of large hydrostatic stress)
- Empirical correlation for densification is a function of
  - β - Burnup (in FIMA)
  - $\Delta\rho_0$ – Total densification that can occur (a common value is 0.01)
  - β_D – Burnup at which densification stops (a common value is 5 MWD/kgU)
  - C_D = 7.235 − 0.0086 (T(° C) − 25) for T < 750° C and CD = 1 for T ≥ 750 ° C

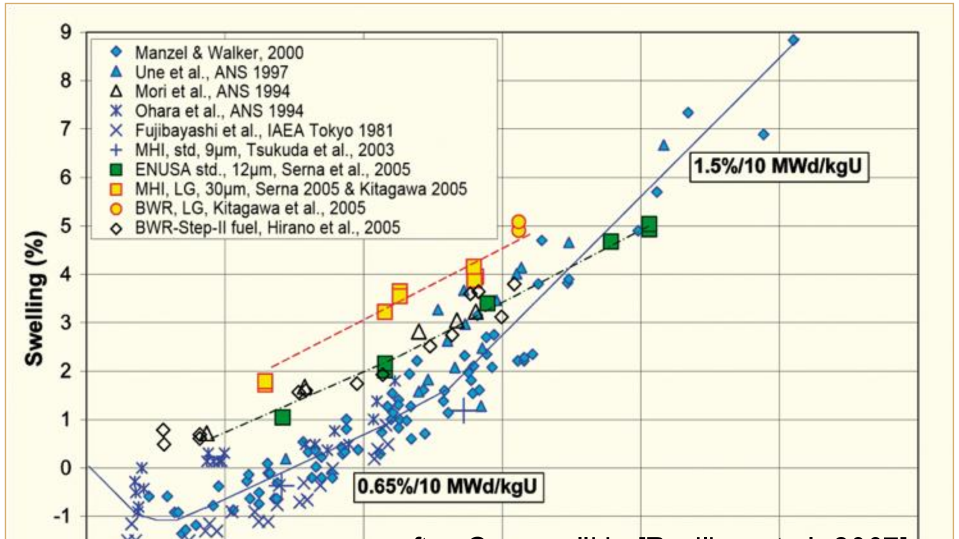
$$\epsilon_D = \Delta\rho_0 \left(e^{\frac{\beta \ln 0.01}{C_D \beta_D}} - 1 \right)$$





Fission product induced swelling

- Fission product swelling results from three changes in the fuel microstructure
  - Solid swelling: Accumulation of soluble and insoluble fission products in fuel matrix
  - Gaseous swelling: Accumulation of gaseous and volatile fission products in intragranular and intergranular pores
  - High burnup swelling: Restructuring of pellet rim with the accumulation of fission gas in a large number of small pores

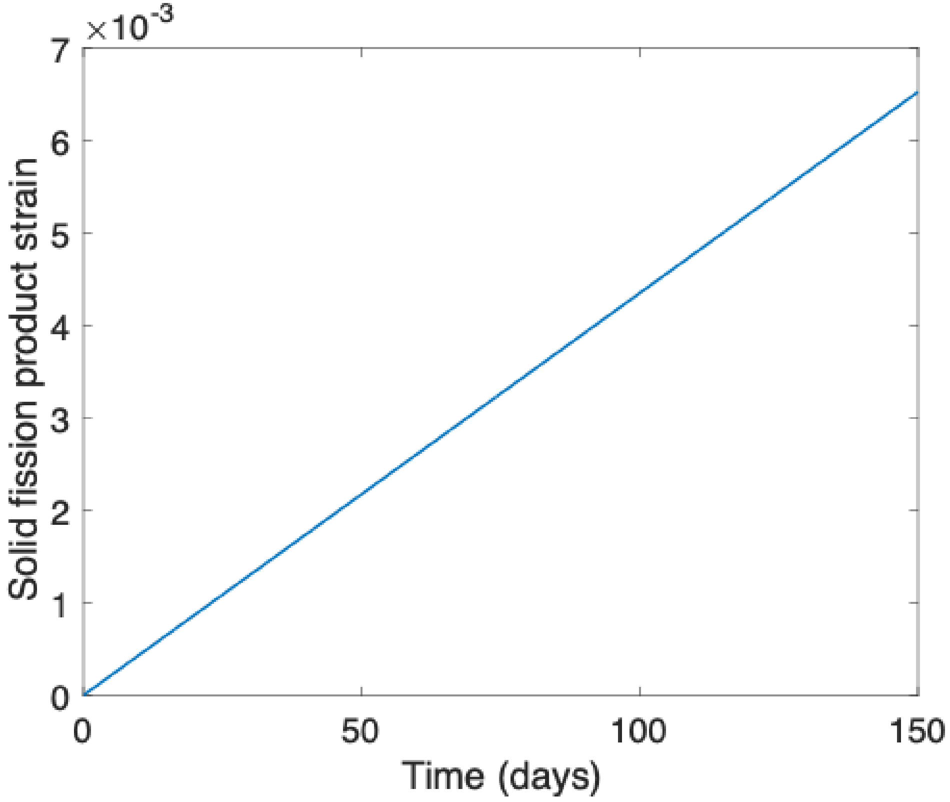


after Garzarolli in [Rudling et al, 2007]

Solid fission product swelling

- The solid fission product swelling model is a function of:
  - β – Burnup (in FIMA)
  - ρ – Initial UO_2 density (g/cm^3)
- Includes contributions from soluble oxides, insoluble oxides, and metallic precipitates

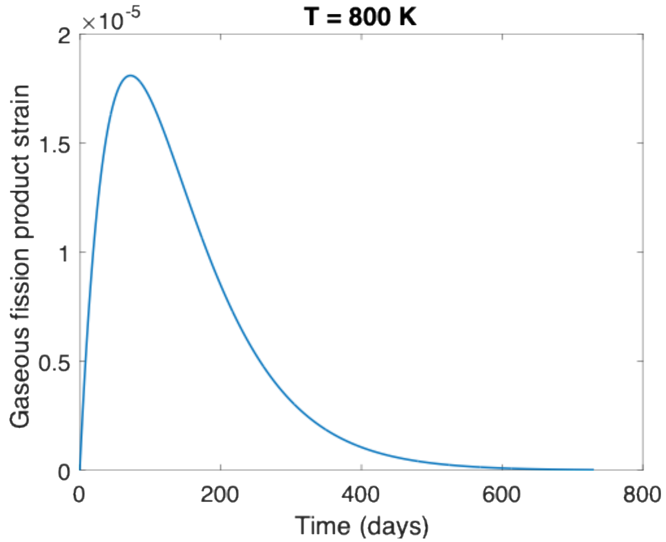
$$\epsilon_{sfp} = 5.577 \times 10^{-2} \rho\beta$$

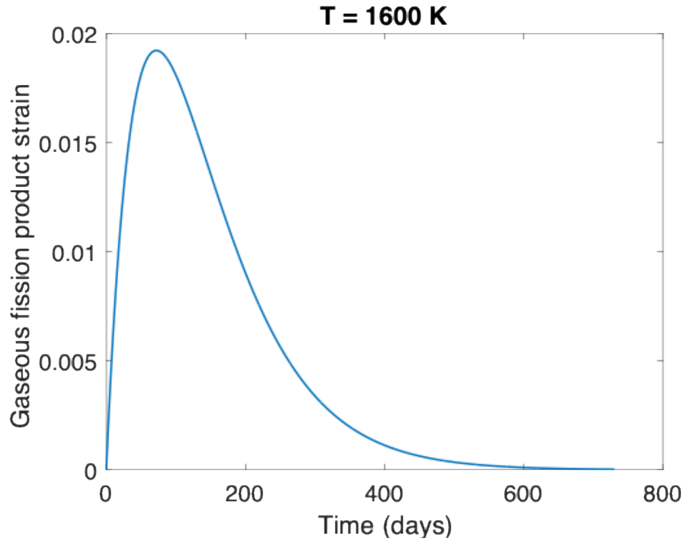


Gaseous fission product swelling

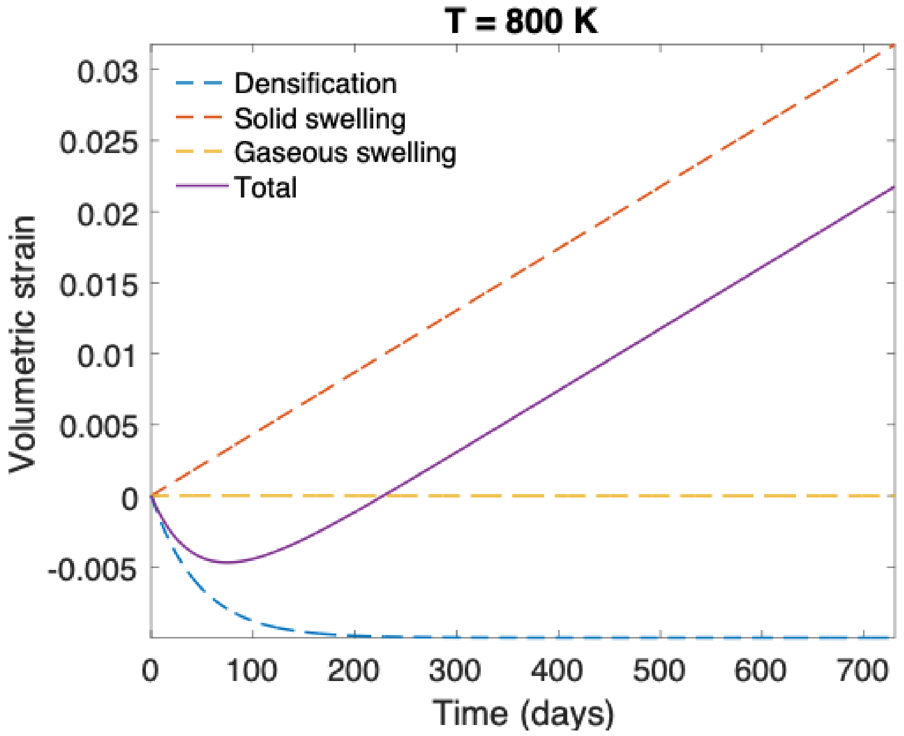
- Gaseous swelling varies strongly with temperature, fission rate, and stress
- Empirical relationships dependent upon burnup and temperature
- T < 1000K
  - Fission gas atoms remain in fuel matrix or collect in small, isolated, intragranular pores (<1 nm)
  - Intragranular pore size limited by fission spikes that drive gas back into fuel matrix
  - Gaseous swelling constrained by fission gas release
- T = 1000 to 1700 K
  - Swelling takes place at hot interior of pellet
  - Gas atoms in fuel matrix diffuse to grain boundaries and collect in pores
  - Gas pressure causes bubbles to increase in size and to coalesce into larger pores
  - Gaseous swelling opposed by applied stress
  - Gaseous swelling also constrained by fission gas release

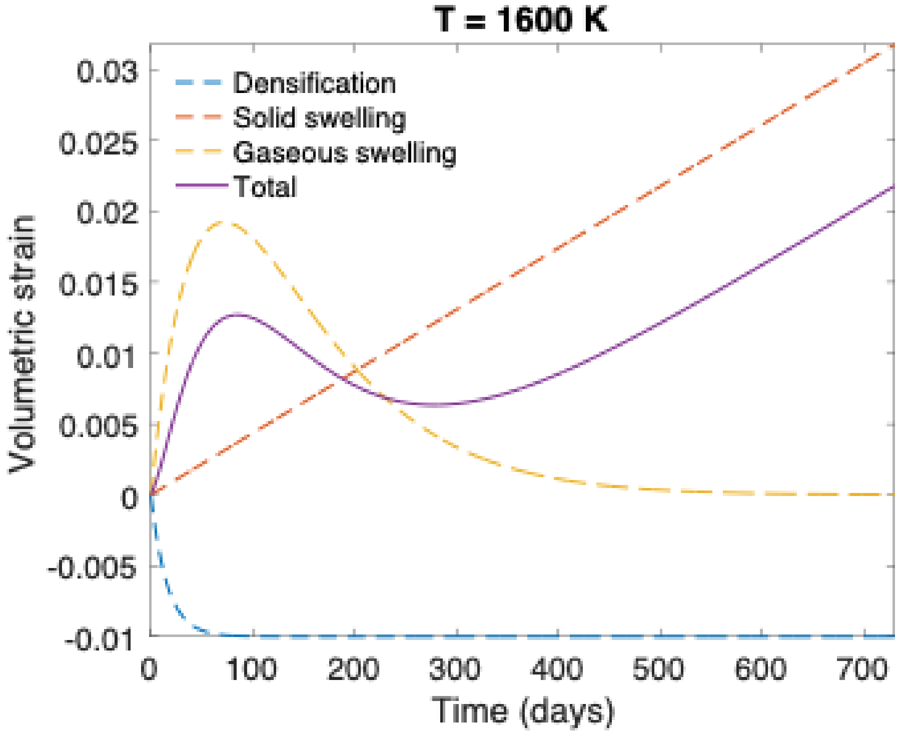
$$\epsilon_{gfp} = 1.96 \times 10^{-28} \rho\beta (2800 - T)^{11.73} e^{-0.0162(2800-T)}\, e^{-17.8\rho\beta}$$





The overall swelling behavior depends on temperature





Total change in volume

- The total change in volume is found by adding all components of dimensional change
  - $\varepsilon_{tot} = \varepsilon_{th} + \varepsilon_D + \varepsilon_{sfp} + \varepsilon_{gfp}$
- Example:
  - fission rate = 2.5e13 f/(cm^3 s)
  - T(fuel) = 1400 K
  - T_{ref} = 300 K
  - For densification: $\Delta\rho_0$ = 0.01 and β_D = 5 MWD/kgU
  - Total time: 2 weeks

Example

- Will do this example in the problem session